

BUSINESS FINANCIAL
MONITORING SYSTEM
FRONTEND & USER INTERFACE
DOCUMENTATION

APPLICATION OVERVIEW

The Business Financial Monitoring System is a desktop application developed using Python. The graphical interface was implemented using CustomTkinter and Tkinter to create a modern and interactive graphical user interface. The system allows users to:

- **Manage business income and expenses**
- **Scan invoices using OCR technology**
- **Generate financial reports**
- **Visualize financial data through charts**
- **Export reports to Excel and CSV**

LOGIN SCREEN

Business Financial Monitoring

LOGIN SCREEN

- The login screen allows users to securely access the system.

Components:

- Username field – enter the user account name
- Password field – enter the secure password
- Login button – authenticates the user
- Register button – creates a new user accoun

Passwords are encrypted using the bcrypt security library.

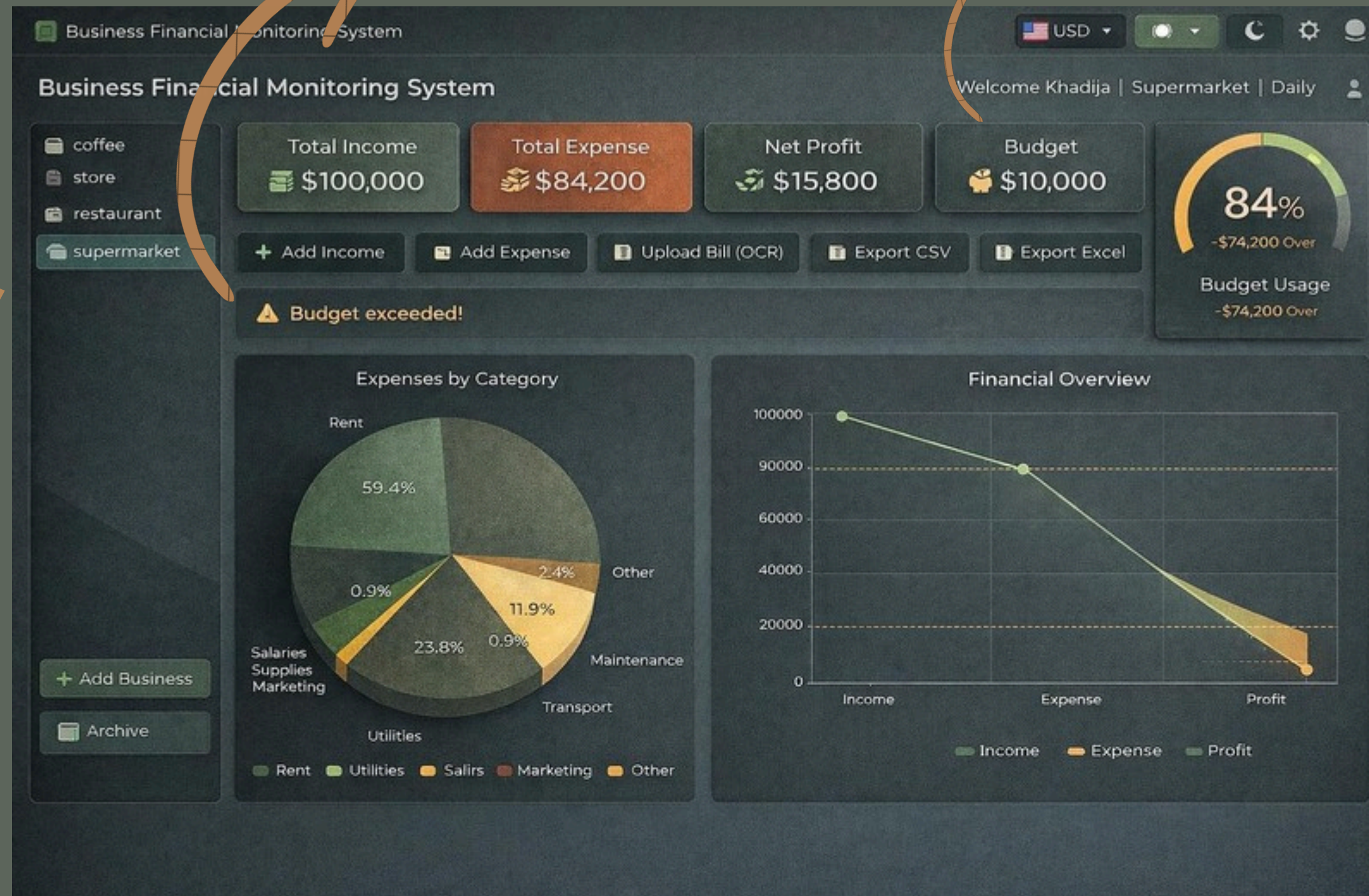
DASHBOARD INTERFACE

Alerts Section

Budget Display

Business List

Add Business Button

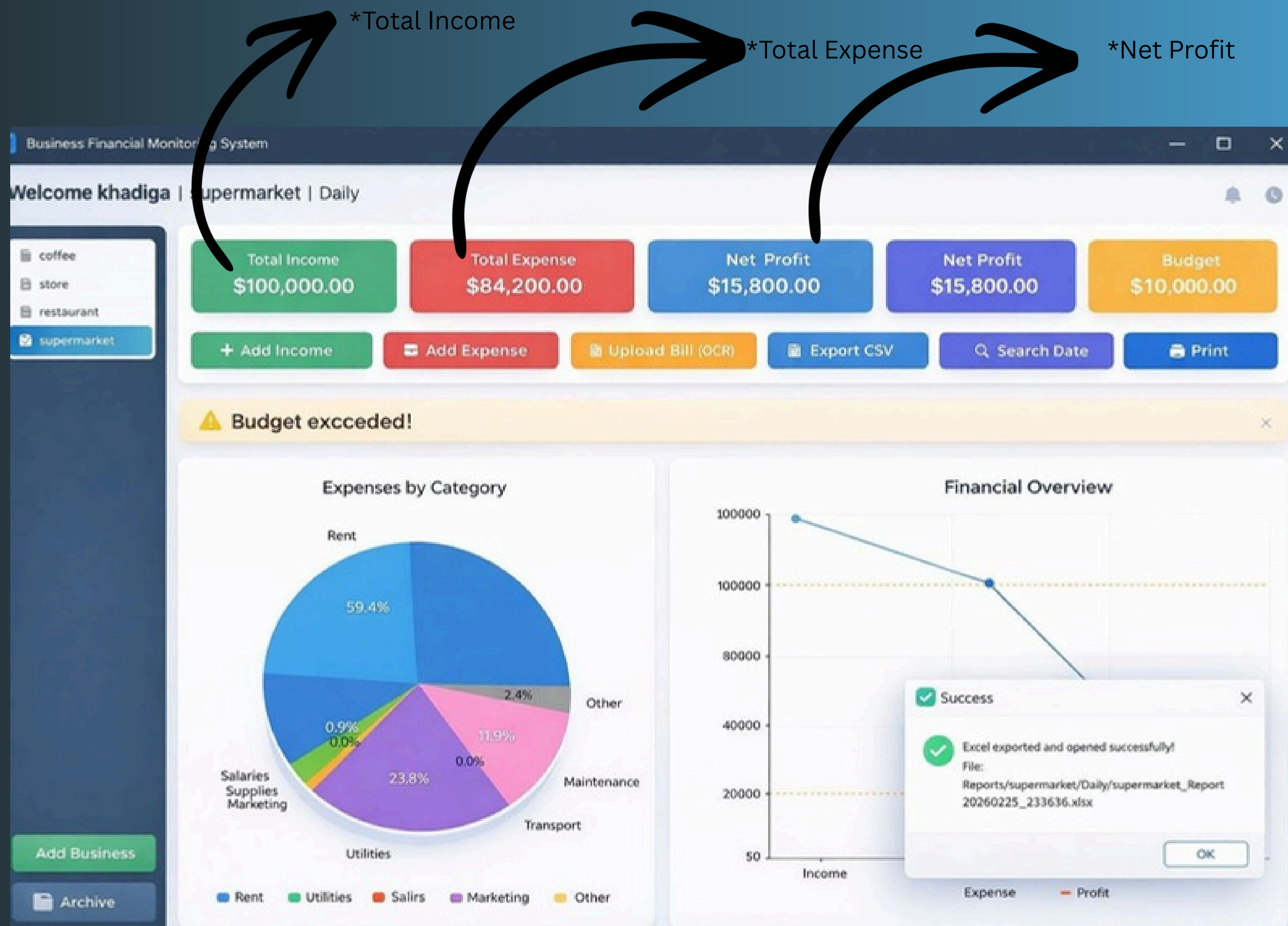


DASHBOARD INTERFACE

The dashboard is the main interface of the application where users manage businesses and monitor financial data.

Components:

- **Business List – displays all businesses created by the user**
- **Add Business Button – creates a new business profile**
- **Budget Display – shows the selected business budget**
- **Alerts Section – displays warnings if expenses exceed the budget**



FINANCIAL SUMMARY SECTION

The financial summary provides a quick overview of the current business financial status. Displayed Information:

- Total Income
- Total Expense
- Net Profit . Profit is calculated using the formula: $\text{Profit} = \text{Income} - \text{Expenses}$

CONTROL BUTTONS

Main control buttons available on the dashboard:

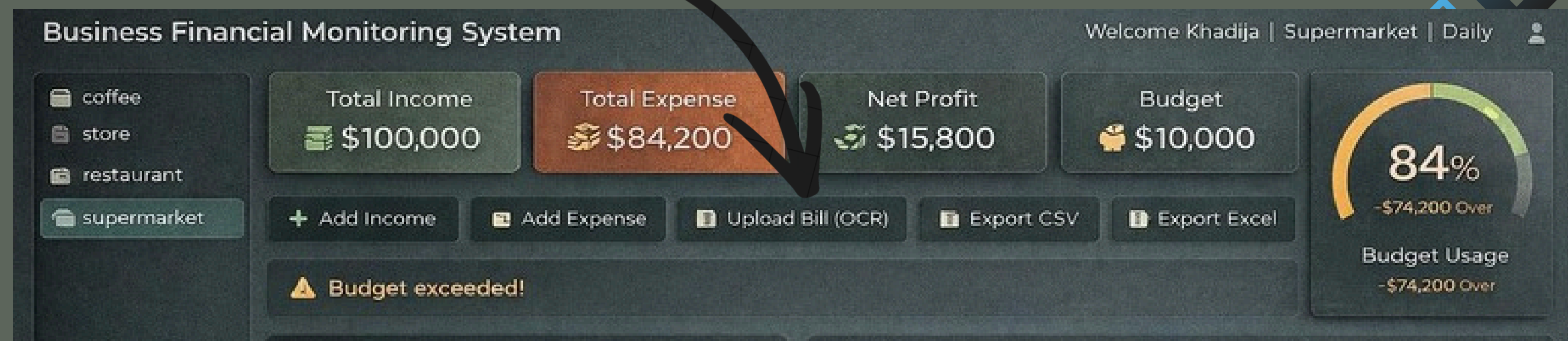
- Add Income – manually add income records
- Add Expense – manually add expense records
- Upload Bill (OCR) – upload an invoice image and extract the amount automatically
- Export CSV – export financial data into a CSV file
- Export Excel – generate a full financial Excel report
- Open Archive – view previously generated reports and notes

OCR INVOICE SCANNER

The system provides an automatic invoice scanning feature.

Process:

1. The user uploads an invoice image.
2. The system processes the image.
3. OCR technology extracts the text from the image.
4. The system detects numbers using regular expressions.
5. The final detected amount is added as an expense record.



FINANCIAL CHARTS

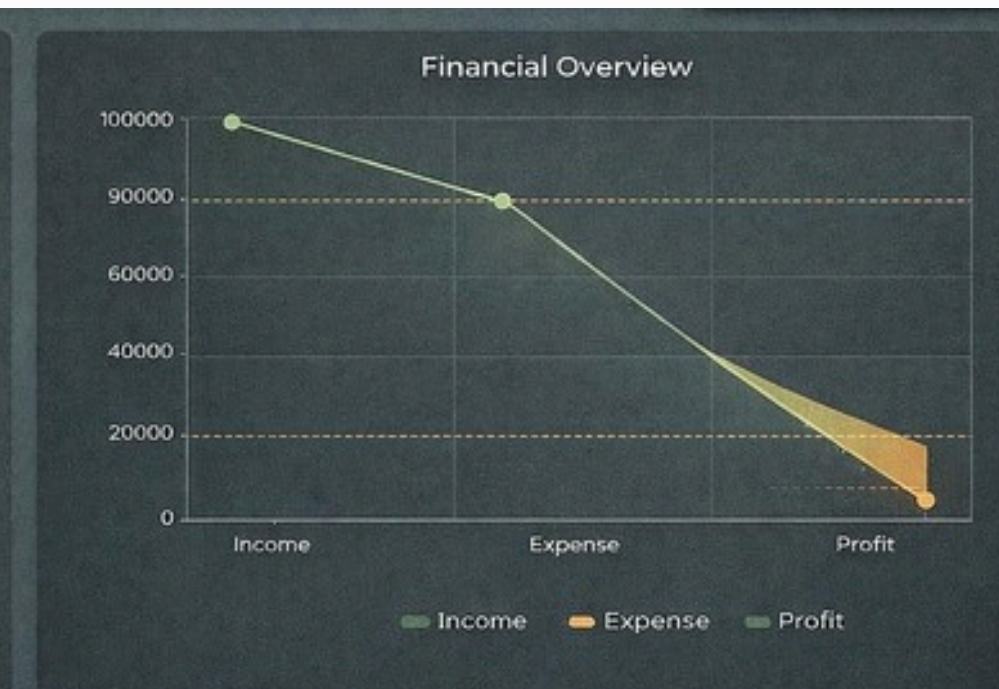
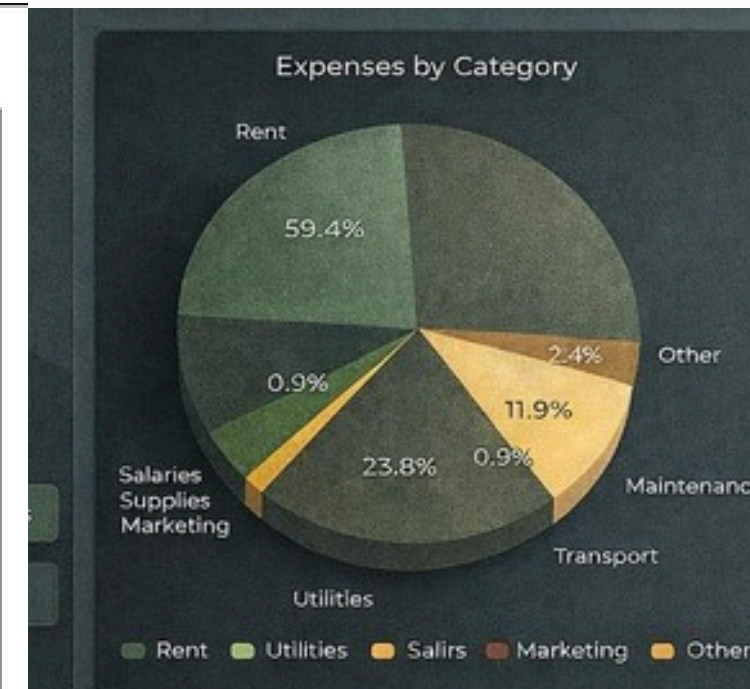
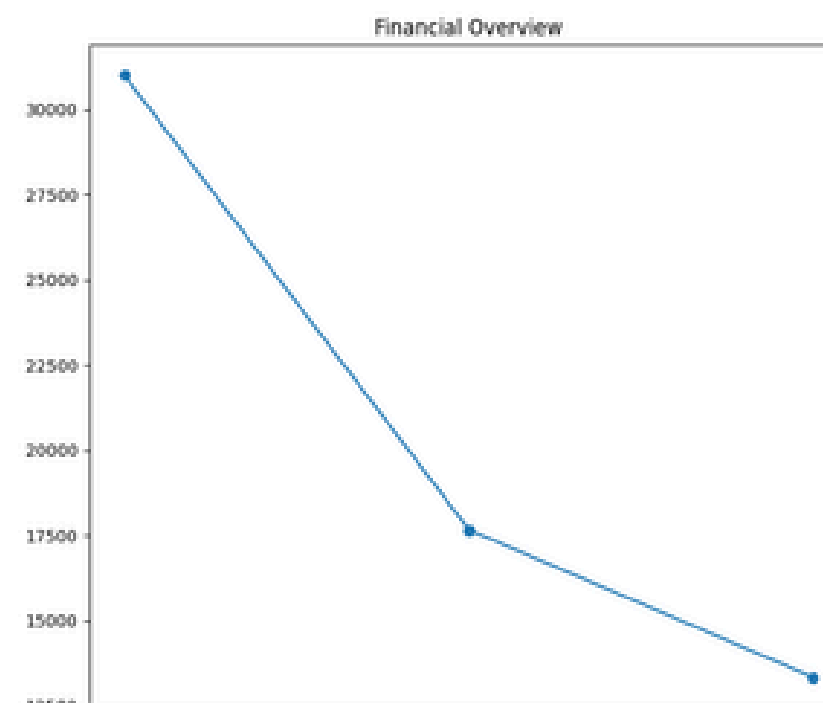
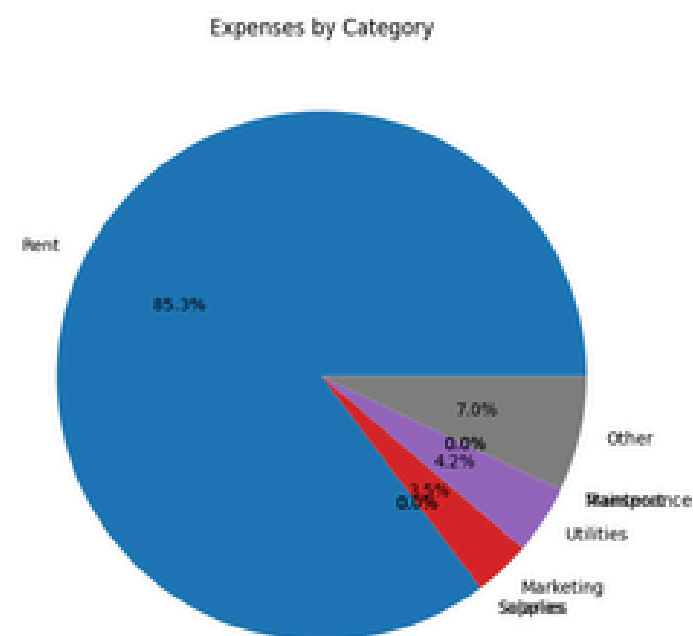
Financial charts are generated using the Matplotlib library. Chart

Types:

- **Pie Chart** – shows the distribution of expenses by category
- **Line Chart** – compares income, expenses, and profit These charts help users visualize financial performanc

Pie Chart

Line Chart



REPORTS AND ARCHIVE

The system automatically generates financial reports.

Features

- **View previous reports**
- **Open reports from the archive**
 - **Add notes to reports**
- **Track financial performance over time**

REPORTS AND ARCHIVE

Business Financial Monitoring System

Welcome None | coffee | Daily

coffee

Store

restaurant

supermarket

My business

Total Income

\$10,000.00

Total Expense

\$50,050.09

Net Profit

\$-40,050.09

Upload Bill (OCR)

Export CSV

Export Excel

Search Date

Report_20260222_135711.xlsx

Report_20260222_135713.xlsx

Report_20260222_135714.xlsx

Report_20260222_135715.xlsx

Report_20260222_135742.xlsx

Report_20260222_135919.xlsx

Report_20260222_135922.xlsx

Report_20260222_142410.xlsx

coffee_Report_20260222_143724.xlsx

coffee_Report_20260224_141438.xlsx

Notes

Save Notes

coffee_Report_20260222_143...

Saved to this PC

Search

Comments

Share

File Home Insert Layout Formulas Data Review View Automate Help

General

Conditional Formatting

Format as Table

Cell Styles

Cells

Editing

Sensitivity

Add-ins

Copilot

Business Name

D E F G H I J K L M N O P Q

1 Business N coffee

2 Period Daily

3 Budget 10000

4 Total Inco 100000

5 Total Expe 9000

6 Net Profit 91000

7

8 TYPE Amount Category Bill Numbe Date

9 Income 100000 - 2026-02-22 14:32:51

10 Expense 1000 Transport 10223 2026-02-22 14:33:35

11 Expense 3000 Other 11102 2026-02-22 14:34:04

12 Expense 5000 Utilities 10002 2026-02-22 14:34:37

13

14

15

16

17

Financial Report

Ready Accessibility: Good to go

Income Expense

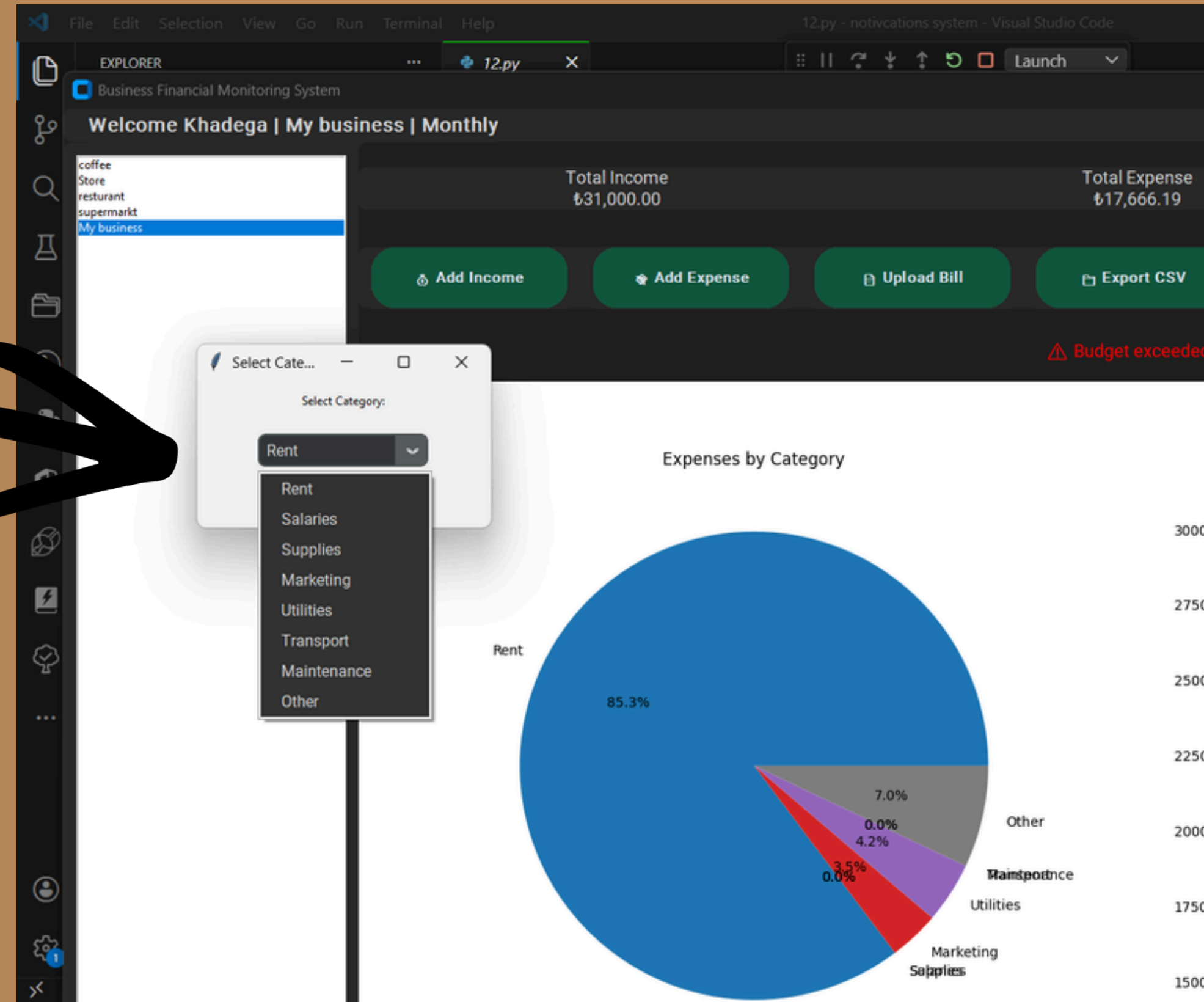
CATEGORIES SYSTEM

Expenses are organized using predefined categories.

Default categories include:

- Rent
- Salaries
- Supplies
- Marketing
- Utilities
- Transport
- Maintenance
- Other

Categories help generate accurate financial analysis and charts



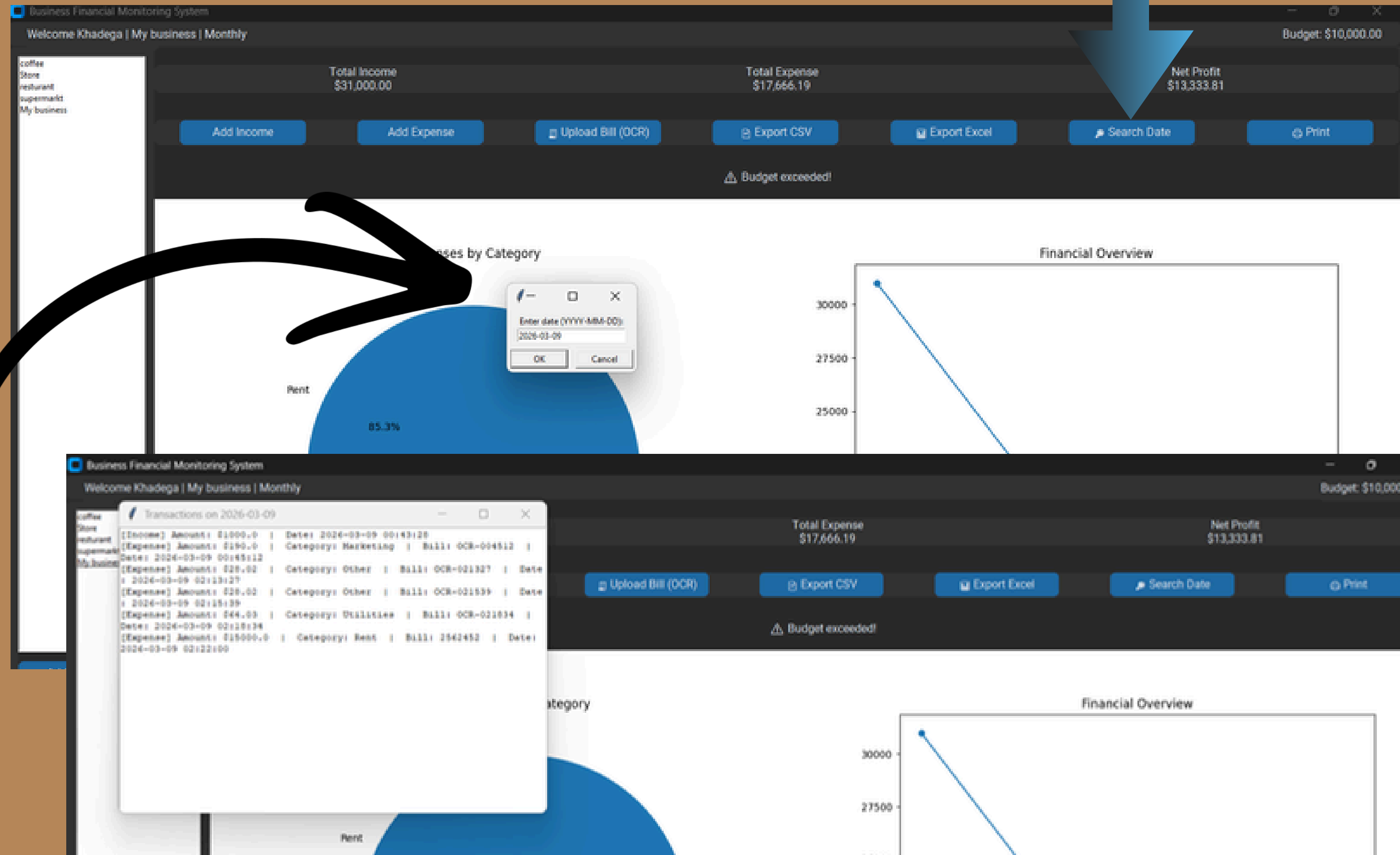
SEARCH FUNCTION

SEARCH ICON

The search feature
allows users to
quickly find financial
records.

Users can search by:

- Invoice number
- Date
- Category
- Amount



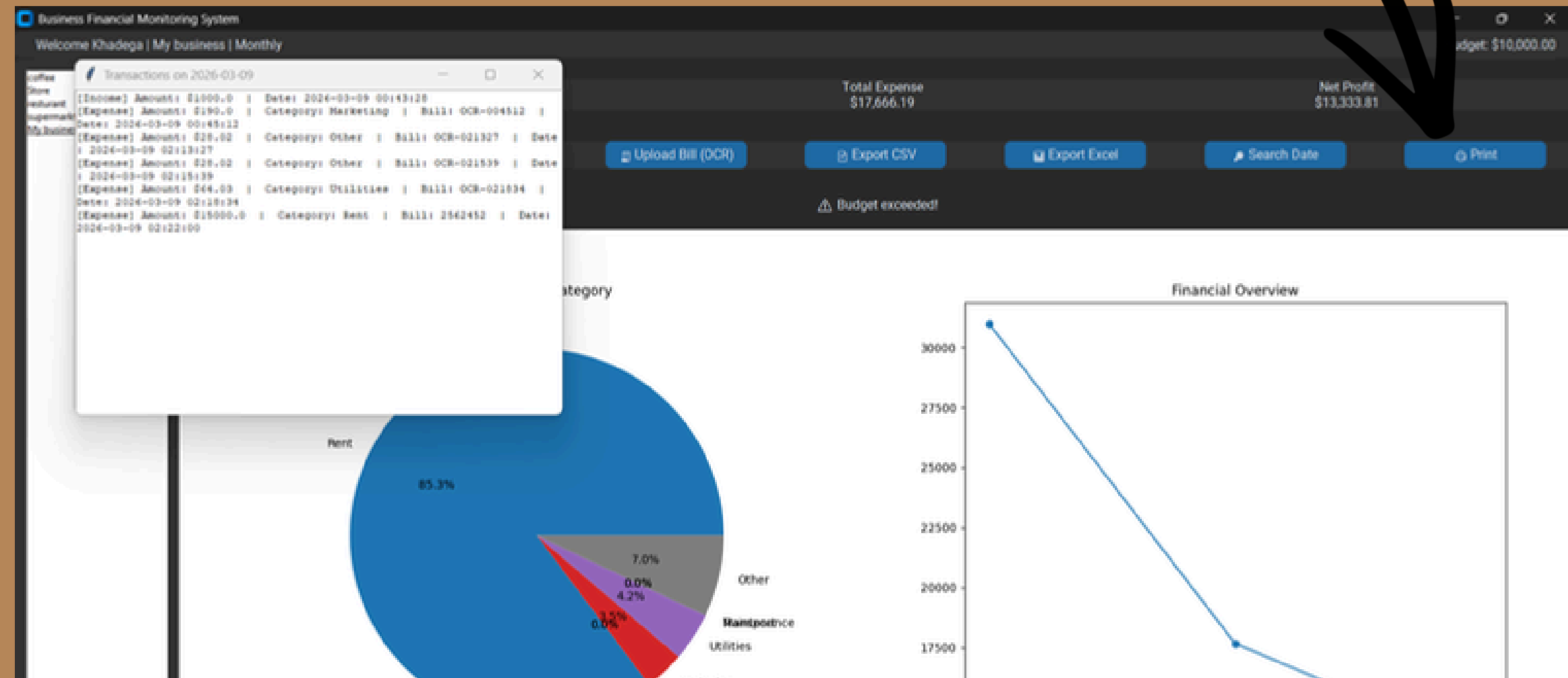
PRINT REPORTS

Users can print financial reports directly

. Process:

- 1. Generate the financial report.**
- 2. Open the Excel report file**
- . 3. Send the report to a printer**

PRINT icon



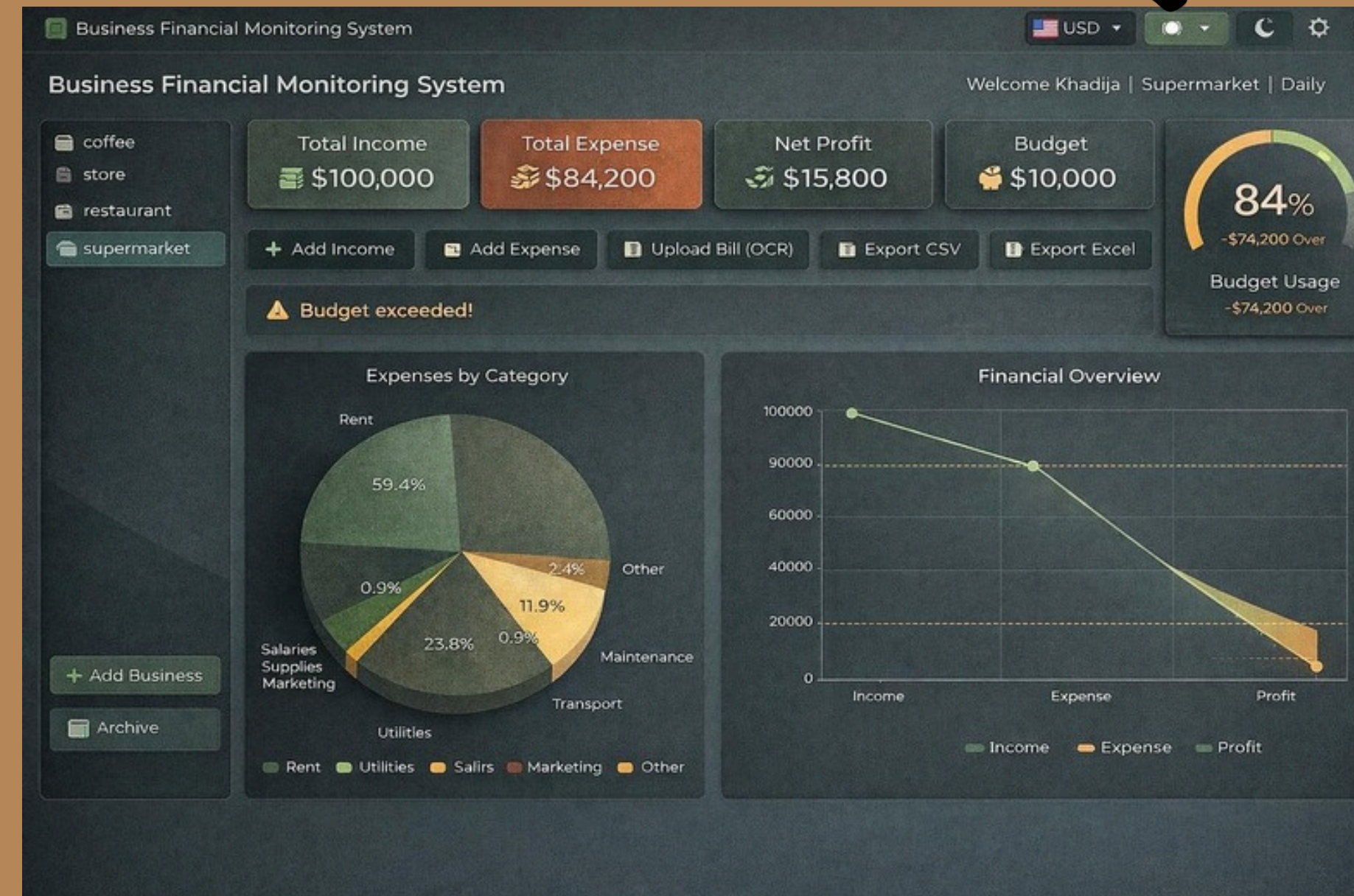
LANGUAGE SELECTION

The application interface supports multiple languages. Supported languages:

- English
- Arabic

Additional languages can be added in future updates.

LANGUAGE SELECTION BUTTON



LIBRARIES USED IN THE PROJECT

Library	Purpose
CustomTkinter	Modern graphical user interface
Tkinter	Core GUI framework
Matplotlib	Financial charts and data visualization
Pillow	Image processing for invoices
Tesseract OCR	Text extraction from invoice images
OpenPyXL	Excel report generation
bcrypt	Password encryption and authentication
csv	Export data to CSV files
json	Store business data
re	Extract numbers using regular expressions

SYSTEM REQUIREMENTS

14. System Requirements

Requirement	Description
Python	Version 3.9 or later
Tesseract OCR	Required for invoice text extraction
Operating System	Windows, macOS, or Linux

HOW TO RUN THE APPLICATION

Step 1 – Install Python (version 3.9 or later)

Step 2 – Download the project from GitHub

Step 3 – Install required libraries: `pip install -r requirements.txt`

Step 4 – Install Tesseract OCR

Step 5 – Run the application using: `python main.py`

System Advantages:

- • **Automated invoice processing using OCR**
- • **Real-time financial monitoring**
- • **Data visualization with charts**
- • **Secure authentication system • Professional financial reporting**

THANK YOU

2026